



ADS-Ω STANDARD INITIATIVE

A Cyber Unveil International, LLC Publication

AUTONOMOUS DECISION SECURITY

ADS-Ω

The standard for securing the decisions autonomous systems make

TRACEABLE · EXPLAINABLE · GOVERNED · CONTAINED

v0.2.1 · Controlled Pre-Standard · Introductory Overview

The Unsecured Layer

Autonomous systems now make consequential decisions — credit approvals, clinical recommendations, infrastructure actions, security responses — at a scale and speed beyond direct human review.

Conventional security secures the data, the model, and the perimeter. None of them secures the determination those systems produce.

ISO/IEC 27001

Secures information. Silent on the decisions information produces.

NIST AI RMF / ISO 42001

Govern and manage AI. Neither specifies decision-level enforcement at runtime.

EU AI Act

Requires risk management and record-keeping. Does not provide the architecture to produce the evidence.

The decision is the unsecured layer.

ADS-Ω secures it.

THE ADS-Ω THESIS

The defensible enterprise will not be measured by how much it has automated.

It will be measured by whether every consequential decision its systems make is traceable, validated, bounded, and auditable.

*Automation without decision security
is liability at scale.*

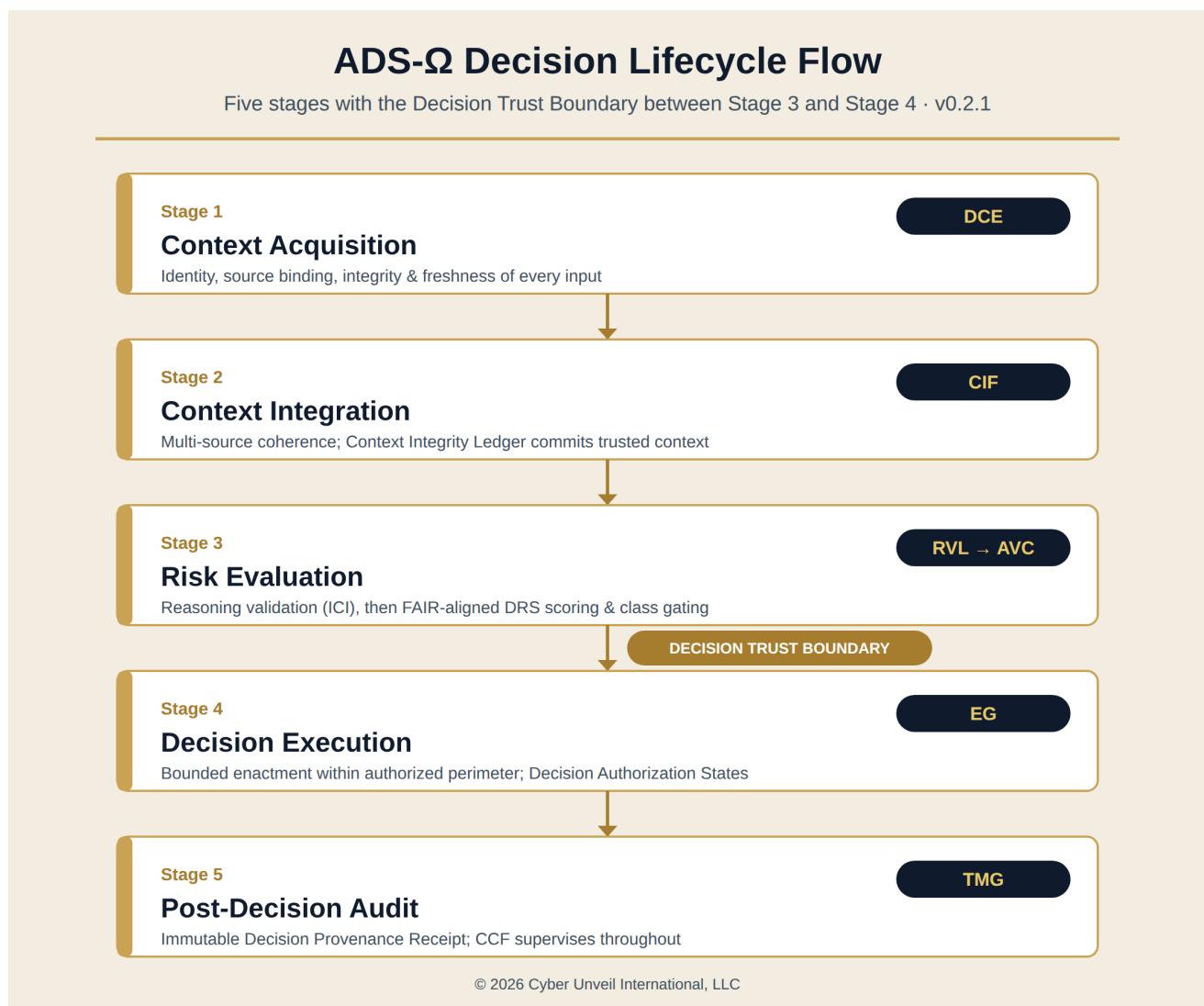
A Seven-Layer Control Plane

ADS-Ω defines seven layers. Each secures one property of a decision. A decision flows through all seven; none is optional in a full implementation.



Five Stages, One Trust Boundary

Every decision traverses five ordered stages. It crosses the Decision Trust Boundary — and executes — only when context, reasoning, risk, and authorization all pass together.



Risk Expressed in Dollars

ADS-Ω quantifies decision risk with a FAIR-aligned Decision Risk Score (DRS) on a 0–100 scale. Boards act on financial exposure, not abstractions.

The DRS model

$$\text{Annual Loss Exposure} = \text{LEF} \times \text{PLM}$$

LEF = Loss Event Frequency

PLM = Probable Loss Magnitude

The Action Vector Control layer assigns every decision to a class — Prohibited, Conditional-Block, Conditional-Grant, Conditional-Suspend — and blocks any decision whose DRS exceeds its class threshold from executing without escalation.

A drifting model that costs \$7.1M in annual exposure scores DRS 72 — and is blocked before it executes.

Three Tiers of Conformance

ADS-Ω is adopted progressively. Even the foundational tier materially improves decision security over unmanaged automation.

FOUNDATIONAL · Tier 1

DCE + AVC + TMG

Decision identity, risk scoring, and immutable trace. Minimum viable decision security.

OPERATIONAL · Tier 2

Adds CIF + RVL + EG

The full pre-execution control plane. The Decision Trust Boundary enforced. Target state for regulated industries.

ADVANCED · Tier 3

Adds CCF

Continuous drift detection, containment, and adversarial validation. For critical infrastructure and high-consequence decisions.



Read the Full Standard

This overview is the public introduction to ADS-Ω. The full v0.2.1 Standard — the complete normative text, all requirements, and the Implementation Technical Guide — is available under controlled access to qualifying organizations.

standards.cyberunveil.com

What qualifying reviewers receive

- ADS-Ω Standard v0.2.1 — full normative text
- Implementation Technical Guide v1.0
- Architectural reference diagrams
- Founding Adopter Program (on request)

*Your enterprise has secured its data.
Has it secured the decisions that data produces?*